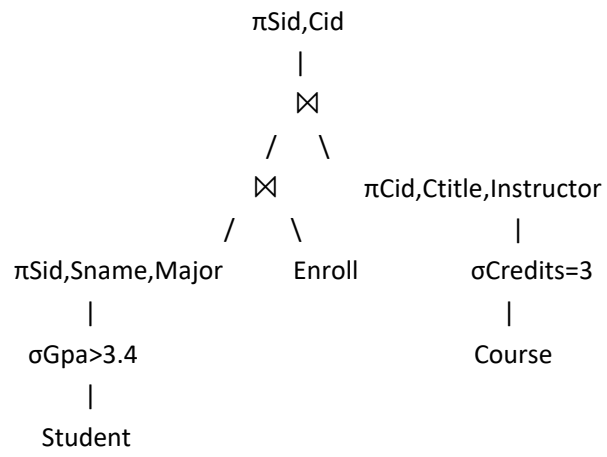


Part2

Problem 1:



Problem 2:

Step1:

Rule 1: Conjunctive selection operations can be deconstructed into a sequence of individual selections.

$$\sigma_{\theta_1 \wedge \theta_2}(E) = \sigma_{\theta_1}(\sigma_{\theta_2}(E))$$

Rule used: Conjunctive selection decomposition (Rule 1)

$$\sigma_{\{\theta_1 \wedge \theta_2\}}(E) = \sigma_{\theta_1}(\sigma_{\theta_2}(E))$$

$$\text{Then } \sigma_{\{\theta_1 \wedge \theta_2\}}(E_1 \bowtie_{\theta_3} E_2) = \sigma_{\theta_1}(\sigma_{\theta_2}(E_1 \bowtie_{\theta_3} E_2))$$

Step2:

Rule 7 (a): When all the attributes in θ_0 involve only the attributes of one of the expressions (E_1) being joined.

$$\sigma_{\theta_0}(E_1 \bowtie_{\theta} E_2) = (\sigma_{\theta_0}(E_1)) \bowtie_{\theta} E_2$$

Rule used: Selection pushdown through join (Rule 7(a))

$$\sigma_{\theta}(E_1 \bowtie_{\theta_3} E_2) = E_1 \bowtie_{\theta_3} (\sigma_{\theta}(E_2)) \text{ when } \theta \text{ involves only attributes of } E_2$$

Since θ_2 involves only attributes of E_2

$$\text{Then } \sigma_{\theta_1}(\sigma_{\theta_2}(E_1 \bowtie_{\theta_3} E_2)) = \sigma_{\theta_1}(E_1 \bowtie_{\theta_3} (\sigma_{\theta_2}(E_2)))$$

Step3:

$$\sigma_{\{\theta_1 \wedge \theta_2\}}(E_1 \bowtie_{\theta_3} E_2) = \sigma_{\theta_1}(E_1 \bowtie_{\theta_3} (\sigma_{\theta_2}(E_2)))$$

Problem3:

$$\text{Selectivity}(\text{Major}=\text{'Math'}) = 1/100$$

$$n(\sigma\{\text{Major}=\text{'Math'}\}(\text{Student})) = 20000 * (1/100) = 200$$

$$\text{Selectivity}(\text{Gpa} > 3.00) = (4.00-3.00)/(4.00-0.00) = 0.25$$

$$n(\sigma\{\text{Gpa} > 3.00\}(\text{Student})) = 20000 * 0.25 = 5000$$

(a)

$$\text{Selectivity}(\text{Major}=\text{'Math'} \wedge (\text{Gpa} > 3.00))(\text{Student})) = (1/100) * 0.25 = 0.0025$$

$$n(\sigma\{(\text{Major}=\text{'Math'}) \wedge (\text{Gpa} > 3.00)\}(\text{Student})) = 20000 * 0.0025 = 50$$

(b)

$$\text{Selectivity}(\theta_1 \vee \theta_2) = \text{Sc}(\theta_1) + \text{Sc}(\theta_2) - \text{Sc}(\theta_1 \wedge \theta_2)$$

$$\text{Selectivity}((\text{Major}=\text{'Math'}) \vee (\text{Gpa} > 3.00)) = \text{Selectivity}(\text{Major}=\text{'Math'}) + \text{Selectivity}(\text{Gpa} > 3.00) -$$

$$\text{Selectivity}((\text{Major}=\text{'Math'}) \wedge (\text{Gpa} > 3.00)) = 0.01 + 0.25 - 0.0025 = 0.2575$$

$$n(\sigma\{(\text{Major}=\text{'Math'}) \vee (\text{Gpa} > 3.00)\}(\text{Student})) = 20000 * 0.2575 = 5150$$

(c)

$$\text{Selectivity}(\text{Major} \neq \text{'Math'}) = 1 - 1/100 = 99/100$$

$$\text{Selectivity}((\text{Major} \neq \text{'Math'}) \wedge (\text{Gpa} > 3.00)) = (99/100) * 0.25 = 0.2475$$

$$n(\sigma\{(\text{Major} \neq \text{'Math'}) \wedge (\text{Gpa} > 3.00)\}(\text{Student})) = 20000 * 0.2475 = 4950$$

Problem4:

Step 1: Push selection down

$$\pi_{\text{Sid}, \text{Dean}} \sigma_{\{\text{Major} = \text{'Math'}\}} (\text{Student} \bowtie \text{Major} = \text{Dcode}(\text{Department} \bowtie \{\text{School} = \text{Scode}\} \text{School})) =$$
$$\pi_{\text{Sid}, \text{Dean}} ((\sigma_{\{\text{Major} = \text{'Math'}\}}(\text{Student})) \bowtie \text{Major} = \text{Dcode}(\text{Department} \bowtie \{\text{School} = \text{Scode}\} \text{School}))$$

Step 2: Push projections down

Optimized Query 1:

$$\pi_{\text{Sid}, \text{Dean}} (\pi_{\text{Sid}, \text{Major}} (\sigma_{\{\text{Major} = \text{'Math'}\}}(\text{Student})) \bowtie \{\text{Major} = \text{Dcode}\} (\pi_{\text{Dcode}, \text{School}} (\text{Department} \bowtie \{\text{School} = \text{Scode}\} \pi_{\text{Scode}, \text{Dean}} (\text{School})))$$

Optimized Query 2:

$$\pi_{\text{Sid}, \text{Dean}} ((\pi_{\text{Sid}, \text{Major}} (\sigma_{\{\text{Major} = \text{'Math'}\}}(\text{Student}))) \bowtie \{\text{Major} = \text{Dcode}\} \pi_{\text{Dcode}, \text{School}} (\text{Department})) \bowtie \{\text{School} = \text{Scode}\} \pi_{\text{Scode}, \text{Dean}} (\text{School})$$

Problem5:

1. Relation sizes and number of blocks

Student:

Tuple size:

$$10+30+10+5 = 55 \text{ bytes}$$

Blocking factor:

$$\text{bfr}(\text{Student}) = \lceil 500/55 \rceil = 9$$

Number of blocks:

$$\text{B}(\text{Student}) = \lceil 20000/9 \rceil = 2223$$

Department:

Tuple size:

$$10+30+10+50+10 = 110 \text{ bytes}$$

Blocking factor:

$$\text{bfr}(\text{Department}) = \lceil 500/110 \rceil = 4$$

Number of blocks:

$$\text{B}(\text{Department}) = \lceil 200/4 \rceil = 50$$

School:

Tuple size:

$$10+30+30+50+100=220 \text{ bytes}$$

Blocking factor:

$$\text{bfr}(\text{School}) = \lceil 500/220 \rceil = 2$$

Number of blocks:

$$\text{B}(\text{School}) = \lceil 20/2 \rceil = 10$$

2. Intermediate result sizes

Because of the foreign-key constraints:

① each Student matches exactly one Department

② each Department matches exactly one School

So:

Student ⋈ Department:

Number of tuples:

$$\text{T}(\text{Student} \bowtie \text{Department}) = 20000$$

Tuple size (duplicate join column kept):

$$55+110 = 165 \text{ bytes}$$

Blocking factor:

$$\text{bfr}(\text{SD}) = \lceil 500/165 \rceil = 3$$

$$\text{Blocks: B}(\text{SD}) = \lceil 20000/3 \rceil = 6667$$

Department ⋈ School:

Number of tuples:

$T(\text{Department} \bowtie \text{School}) = 200$

Tuple size:

$110 + 220 = 330$ bytes

Blocking factor:

$bfr(DS) = \lceil 500/330 \rceil = 2$

Blocks: $B(DS) = 200$

Final result:

Either way, final join result has:

$T(\text{final}) = 20000$

Tuple size:

$55 + 110 + 220 = 385$ bytes

Blocking factor:

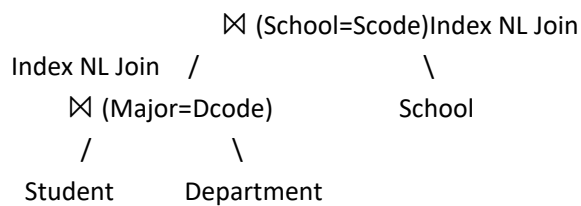
$bfr(\text{final}) = \lceil 500/385 \rceil = 2$

Blocks:

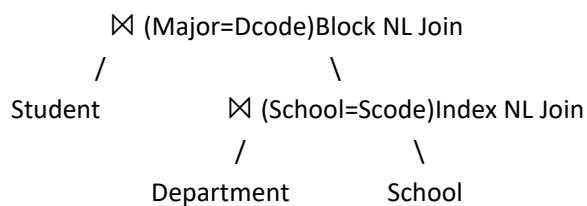
$B(\text{final}) = 20000$

3. Three evaluation plans

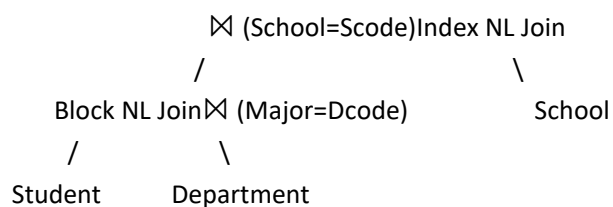
Plan1:



Plan2:



Plan3:



Plan 1

First do Student $\bowtie$ Department using index nested-loop join, then join with School using index

nested-loop join.

$(\text{Student} \bowtie \{\text{Major}=\text{Dcode}\}\text{Department}) \bowtie \{\text{School}=\text{Scode}\}\text{School}$

Step 1:

scan outer Student once: $B(\text{Student}) = 2223$

for each Student tuple, probe Department index:

index height = 4

then 1 more block to fetch matching Department tuple

So per Student tuple:

$4+1 = 5$

Total index lookup cost:

$20000 \times 5 = 100000$

write intermediate SD: $B(\text{SD}) = 6667$

So step 1 cost:

$2223 + 100000 + 6667 = 108890$

Step 2:

$\text{SD} \bowtie \text{School}$

Use SD as outer, and use the primary index on School.Scode as inner.

Cost:

scan outer SD: $B(\text{SD}) = 6667$

for each SD tuple, probe School index:

index height = 2

plus 1 data block

So per tuple:

$2+1=3$

Total lookup cost:

$20000 \times 3 = 60000$

write final result:

$B(\text{final}) = 20000$

So step 2 cost:

$6667 + 60000 + 20000 = 86667$

Total cost of Plan 1

$108890 + 86667 = 195557$

Plan 2:

First do $\text{Department} \bowtie \text{School}$ using index nested-loop join, then join the result with Student using block nested-loop join.

$\text{Student} \bowtie \{\text{Major}=\text{Dcode}\}(\text{Department} \bowtie \{\text{School}=\text{Scode}\}\text{School})$

To reduce cost in the second join, use the smaller relation DS as the outer.

Step 1:

$\text{Department} \bowtie \text{School}$

Use Department as outer, School as indexed inner.

Cost:

scan Department once:

$$B(\text{Department}) = 50$$

for each Department tuple, probe School index:

height 2

plus 1 data block

Per tuple:

$$2+1 = 3$$

$$\text{Total lookup cost: } 200 \times 3 = 600$$

write intermediate DS:

$$B(\text{DS}) = 200$$

So step 1 cost:

$$50+600+200 = 850$$

Step 2:

$\text{Student} \bowtie \text{DS}$

Use block nested-loop join, with DS as outer and Student as inner.

Cost formula:

$$B(\text{outer})+B(\text{outer}) \cdot B(\text{inner})+B(\text{output})$$

So:

$$200+200 \cdot 2223+20000 = 200+444600+20000 = 464800$$

Total cost of Plan 2

$$850+464800 = 465650$$

Plan 3:

First do $\text{Student} \bowtie \text{Department}$ using block nested-loop join, then join the result with School using index nested-loop join.

$$(\text{Student} \bowtie \{\text{Major}=\text{Dcode}\}\text{Department}) \bowtie \{\text{School}=\text{Scode}\}\text{School}$$

For the first join, use the smaller relation Department as outer.

Step 1:

$\text{Department} \bowtie \text{Student}$

Use Department as outer, Student as inner in block nested-loop join.

$$\begin{aligned} \text{Cost: } B(\text{Department})+B(\text{Department}) \cdot B(\text{Student})+B(\text{SD}) &= 50+50 \cdot 2223+6667 = 50+111150+6667 \\ &= 117867 \end{aligned}$$

Step 2:

$\text{SD} \bowtie \text{School}$

Use index nested-loop join with SD as outer and School as indexed inner.

As computed before:

$$6667+20000 \cdot 3+20000 = 6667+60000+20000 = 86667$$

Total cost of Plan 3

$$117867+86667 = 204534$$

Among these three plans: Plan 1 is the cheapest. because both joins can exploit the available primary indexes on: Department.Dcode and School.Scode.